

MATH3018 Assignment 2, due on Feb 12

1. Let $A = \begin{pmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{pmatrix}$.

(a) Calculate $AA^\top$ and obtain its eigenvalues and eigenvectors. [2pt]

(b) Calculate $A^\top A$ and obtain its eigenvalues and eigenvectors. [2pt]

(c) Obtain the singular value decomposition of A . [1pt]

Solution:

(a) We find

$$AA^\top = \begin{pmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{pmatrix} \begin{pmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{pmatrix} = \begin{pmatrix} 25 & 50 & 5 \\ 50 & 100 & 10 \\ 5 & 10 & 145 \end{pmatrix}$$

To find eigenvalues, solve $|AA^\top - \lambda I| = 0$:

$$\begin{vmatrix} 25 - \lambda & 50 & 5 \\ 50 & 100 - \lambda & 10 \\ 5 & 10 & 145 - \lambda \end{vmatrix} = 0$$

The characteristic equation gives us eigenvalues: $\lambda_1 = 150, \lambda_2 = 120, \lambda_3 = 0$

The corresponding eigenvectors are:

$$u_1 = \frac{1}{\sqrt{30}} \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, u_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, u_3 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

(b) Now calculate $A^\top A$:

$$A^\top A = \begin{pmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{pmatrix} \begin{pmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{pmatrix} = \begin{pmatrix} 144 & -12 \\ -12 & 126 \end{pmatrix}$$

The eigenvalues are $\lambda_1 = 150, \lambda_2 = 120$

The corresponding eigenvectors are:

$$v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}, v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

(c) The singular value decomposition of A is $A = U\Sigma V^\top$ where:

$$U = \begin{pmatrix} 1/\sqrt{30} & -1/\sqrt{6} & -2/\sqrt{5} \\ 2/\sqrt{30} & -2/\sqrt{6} & 1/\sqrt{5} \\ 5/\sqrt{30} & 1/\sqrt{6} & 0 \end{pmatrix} \quad \Sigma = \begin{pmatrix} \sqrt{150} & 0 \\ 0 & \sqrt{120} \\ 0 & 0 \end{pmatrix} \quad V = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}$$

or written as

$$U = \begin{pmatrix} 1/\sqrt{30} & -1/\sqrt{6} \\ 2/\sqrt{30} & -2/\sqrt{6} \\ 5/\sqrt{30} & 1/\sqrt{6} \end{pmatrix} \quad \Sigma = \begin{pmatrix} \sqrt{150} & 0 \\ 0 & \sqrt{120} \end{pmatrix} \quad V = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}$$

2. Show that every eigenvalue of a $k \times k$ positive definite matrix A is positive. [2pt]
(Hint: Consider the definition of an eigenvalue, where $Au = \lambda u$. Multiply on the left by $u^\top$.)

Solution: Let λ be an eigenvalue of A and u be the associated eigenvector. By definition $Au = \lambda u$. Multiply both sides of this equation on the left by $u^\top$, we have $\lambda u^\top u = u^\top Au > 0$ where we use the positive-definiteness of A . Since $u^\top u$ is always positive, we conclude $\lambda > 0$.

3. For any positive semi-definite matrix A , let $\lambda_1(A)$ denote its largest eigenvalue. Show that $x^\top Ax \leq \lambda_1(A)$ for any unit vector x . For which x does equality holds? [2pt]

Solution: Since A is positive semi-definite, denote its dimension as n , and we denote the eigen-decomposition to be $A = U\Lambda U^\top$ where Λ is a diagonal matrix consisting of the eigenvalues of A satisfying $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$, and $U = (u_1, u_2, \dots, u_n)$ consists of the eigenvectors of $A \in \mathbb{R}^{n \times n}$.

Then for any unit vector $x \in \mathbb{R}^n$, c_i^2 are non-negative with $\sum_{i=1}^n c_i^2 = 1$:

$$x^\top Ax = \sum_{i=1}^n \lambda_i (x^\top u_i)^2 \leq \lambda_1 \sum_{i=1}^n (x^\top u_i)^2 = \lambda_1$$

where in the second step we use $\lambda_i \leq \lambda_1$ for all i and in the last step we use $\sum_{i=1}^n (x^\top u_i)^2 = x^\top U U^\top x = x^\top x = 1$. Therefore, $x^\top Ax \leq \lambda_1(A)$ for any unit vector x .

Equality holds when $\sum_{i=1}^n \lambda_i (x^\top u_i)^2 = \lambda_1$. Taking $x = \pm u_1$ can achieve this. (If λ_1 is a single eigenvalue, i.e., $\lambda_2 < \lambda_1$, then equality holds if and only if $x = \pm u_1$.)

4. Show that

$$\text{Cov}(c_{11}X_1 + c_{12}X_2 + \dots + c_{1p}X_p, c_{21}X_1 + c_{22}X_2 + \dots + c_{2p}X_p) = \mathbf{c}_1^\top \Sigma \mathbf{c}_2$$

where $\mathbf{c}_1^\top = [c_{11}, c_{12}, \dots, c_{1p}]$, $\mathbf{c}_2^\top = [c_{21}, c_{22}, \dots, c_{2p}]$ and Σ is the population covariance matrix of $\mathbf{X} = (X_1, X_2, \dots, X_p)^\top$. [1.5pt]

Solution: The product

$$\begin{aligned}
& (c_{11}(X_1 - \mu_1) + c_{12}(X_2 - \mu_2) + \cdots \\
& \quad + c_{1p}(X_p - \mu_p))(c_{21}(X_1 - \mu_1) + c_{22}(X_2 - \mu_2) + \cdots + c_{2p}(X_p - \mu_p)) \\
&= \left(\sum_{\ell=1}^p c_{1\ell}(X_\ell - \mu_\ell) \right) \left(\sum_{m=1}^p c_{2m}(X_m - \mu_m) \right) \\
&= \sum_{\ell=1}^p \sum_{m=1}^p c_{1\ell} c_{2m} (X_\ell - \mu_\ell)(X_m - \mu_m)
\end{aligned}$$

has expected value

$$\sum_{\ell=1}^p \sum_{m=1}^p c_{1\ell} c_{2m} \sigma_{\ell m} = [c_{11}, \dots, c_{1p}] \Sigma [c_{21}, \dots, c_{2p}]'.$$

Verify the last step by the definition of matrix multiplication. The same steps hold for all elements.

5. Let $X = (X_1, X_2)^T$ be a random vector. We are given $n = 3$ observations:

$$\mathbf{X} = \begin{pmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{pmatrix}$$

Define $b = (2, 3)^T$ and $c = (-1, 2)^T$. Find the following:

- (i) sample means of $b^T X$ and $c^T X$ [1pt]
- (ii) sample variances of $b^T X$ and $c^T X$, respectively [1pt]
- (iii) sample covariance of $b^T X$ and $c^T X$ [0.5pt]

Solution

Sample mean of X : $\bar{x} = (5, 2)^T$. Sample covariance matrix of X :

$$S = \begin{pmatrix} 16 & -2 \\ -2 & 1 \end{pmatrix}.$$

Sample mean of $b^T X$: $b^T \bar{x} = 16$. Sample mean of $c^T X$: $c^T \bar{x} = -1$. Sample variance of $b^T X$: $b^T S b = 49$. Sample variance of $c^T X$: $c^T S c = 28$. Sample covariance of $b^T X$ and $c^T X$: $b^T S c = -28$.

6. Suppose the random vector $X = (X_1, X_2, X_3)^\top$ has covariance matrix

$$\Sigma = \begin{pmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{pmatrix}.$$

(a) Find the population correlation matrix of X . [1pt]

(b) Find the covariance matrix of the random vector $(X_2 - 3, 2X_1 - X_2 + X_3 + 1)^\top$. [1pt]

Solution

(a)

$$\rho = \begin{pmatrix} 1/5 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{pmatrix} \begin{pmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{pmatrix} \begin{pmatrix} 1/5 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{pmatrix} = \begin{pmatrix} 1 & -1/5 & 4/15 \\ -1/5 & 1 & 1/6 \\ 4/15 & 1/6 & 1 \end{pmatrix}.$$

(b) Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

The covariance matrix of $AX + b$ where $b = (-3, 1)^\top$ is

$$A\Sigma A^T = \begin{pmatrix} 4 & -7 \\ -7 & 135 \end{pmatrix}.$$